WORKED SOLUTION

STEP II 2023, Question 7

Pure • Sums of Two Squares

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 1.6, *Sums of Two Squares*: https://beyondstep.keithmcnulty.org/ch01/two-squares.html

The key idea

A sum of two squares $x^2 + y^2$ is $|x + y\mathrm{i}|^2$. Part (i) then says that the product of two sums of two squares is again a sum of two squares, and multiplying the complex numbers tells you which squares. Every later part works the same way: write each factor as a sum of two squares (often by spotting a nearby perfect square) and multiply. The last part runs this backwards, dividing instead of multiplying.

Solution

Part (i)

We have $zw = (a + b\mathrm{i})(c + d\mathrm{i}) = (ac - bd) + (ad + bc)\mathrm{i}$, so

$$\begin{aligned} |zw|^2 &= (ac - bd)^2 + (ad + bc)^2 \\ &= a^2c^2 - 2abcd + b^2d^2 + a^2d^2 + 2abcd + b^2c^2 \\ &= a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2 \\ &= (a^2 + b^2)(c^2 + d^2) = |z|^2|w|^2. \end{aligned}$$

Both $|zw|$ and $|z||w|$ are non-negative, so taking square roots gives $|zw| = |z||w|$, as required.

In what follows we use the squared form: if z and w have integer real and imaginary parts, then

$$|z|^2\,|w|^2 = |zw|^2 = (\mathrm{Re}\, zw)^2 + (\mathrm{Im}\, zw)^2.$$

Part (ii)

Here $|2+\mathrm{i}|^2 = 4+1 = 5$ and $|10+11\mathrm{i}|^2 = 100+121 = 221$. The product is

$$(2+\mathrm{i})(10+11\mathrm{i}) = 20+22\mathrm{i}+10\mathrm{i}-11 = 9+32\mathrm{i},$$

so by part (i), $9^2+32^2 = 5\times 221$. (Indeed $81+1024 = 1105$.) So $h=9$, $k=32$ works.

Using $10-11\mathrm{i}$ instead (which has the same modulus) gives $(2+\mathrm{i})(10-11\mathrm{i}) = 31-12\mathrm{i}$, so $h=31$, $k=12$ also works.

Part (iii)

The number ends in 5, so $8045 = 5\times 1609$, and $5 = |2+\mathrm{i}|^2$ as before. We now need 1609 as a sum of two squares. It is just above $40^2 = 1600$, and $1609-1600 = 9 = 3^2$, so $1609 = |40+3\mathrm{i}|^2$. Then

$$(2+\mathrm{i})(40+3\mathrm{i}) = 80+6\mathrm{i}+40\mathrm{i}-3 = 77+46\mathrm{i},$$

and part (i) gives

$$77^2+46^2 = 5\times 1609 = 8045.$$

So $m=77$, $n=46$. (Using $3+40\mathrm{i}$ instead gives $-34+83\mathrm{i}$, so $34^2+83^2 = 8045$ too.)

Part (iv)

Here the extra factor is $36 = 6^2 = |6|^2$, the modulus squared of the real number 6. Multiplying $102+201\mathrm{i}$ by 6 simply scales both parts:

$$(6\times 102)^2+(6\times 201)^2 = 36\,(102^2+201^2) = 36\times 50805.$$

So $p=612$, $q=1206$.

Part (v)

The idea is to find two different ways of writing 25 as $|\alpha|^2$ with Gaussian integers, and multiply each by one fixed representation of 1002082.

First, 1002082 is just above $1001^2 = 1002001$, and

$$1002082-1002001 = 81 = 9^2,$$

so $1002082 = 1001^2+9^2 = |1001+9\mathrm{i}|^2$.

Next, $25 = |5|^2 = |3+4\mathrm{i}|^2 = |3-4\mathrm{i}|^2$. Each of these, multiplied by $1001+9\mathrm{i}$, gives a pair:

$$\begin{aligned}
5(1001+9\mathrm{i}) &= 5005+45\mathrm{i},\\
(3+4\mathrm{i})(1001+9\mathrm{i}) &= 3003+27\mathrm{i}+4004\mathrm{i}-36 = 2967+4031\mathrm{i},\\
(3-4\mathrm{i})(1001+9\mathrm{i}) &= 3003+27\mathrm{i}-4004\mathrm{i}+36 = 3039-3977\mathrm{i}.
\end{aligned}$$

Each has modulus squared 25×1002082, by part (i). Taking absolute values of the parts and putting the smaller one first, three distinct pairs with $r < s$ are

$$(r, s) = (45, 5005), \qquad (2967, 4031), \qquad (3039, 3977).$$

Ask yourself Why not use $4 + 3\mathrm{i}$ as well, to get a fourth pair?

Because $4 + 3\mathrm{i} = \mathrm{i}(3 - 4\mathrm{i})$, and multiplying by i just swaps the real and imaginary parts (up to sign). It reproduces $(3039, 3977)$ with the order reversed, so it gives nothing new. The genuinely different choices are 5, $3 + 4\mathrm{i}$ and $3 - 4\mathrm{i}$.

Part (vi)

Now we run the method backwards. Spot a nearby square again: $1002037 - 1002001 = 36 = 6^2$, so

$$1002037 = 1001^2 + 6^2 = |1001 + 6\mathrm{i}|^2,$$

and also $109 = 10^2 + 3^2 = |10 + 3\mathrm{i}|^2$. If we can write $1001 + 6\mathrm{i} = (10 + 3\mathrm{i})(t + u\mathrm{i})$ with t, u integers, then part (i) gives

$$1002037 = 109\,(t^2 + u^2),$$

and dividing by 109 (using $109 \times 9193 = 1002037$) gives $t^2 + u^2 = 9193$.

So we divide, multiplying top and bottom by the conjugate $10 - 3\mathrm{i}$:

$$\frac{1001 + 6\mathrm{i}}{10 + 3\mathrm{i}} = \frac{(1001 + 6\mathrm{i})(10 - 3\mathrm{i})}{109} = \frac{10010 - 3003\mathrm{i} + 60\mathrm{i} + 18}{109}$$
$$= \frac{10028 - 2943\mathrm{i}}{109} = 92 - 27\mathrm{i},$$

since $109 \times 92 = 10028$ and $109 \times 27 = 2943$. The division comes out exactly, so $1001 + 6\mathrm{i} = (10 + 3\mathrm{i})(92 - 27\mathrm{i})$, and therefore

$$92^2 + 27^2 = 9193.$$

So $t = 92$, $u = 27$. As a check, $8464 + 729 = 9193$.

Ask yourself What if the division hadn't come out exactly?

Then we would try the other conjugate. Dividing by $10 - 3\mathrm{i}$ instead gives $(9992 + 3063\mathrm{i})/109$, which is not a Gaussian integer (9992 is not a multiple of 109). Since 109 is prime, one of the two choices must work; this is the Gaussian-integer fact mentioned in Beyond STEP below. Dividing $1001 - 6\mathrm{i}$ gives nothing new, just the conjugates of these.

Checks

Each answer can be checked by direct arithmetic: $9^2 + 32^2 = 1105 = 5 \times 221$; $77^2 + 46^2 = 5929 + 2116 = 8045$; $612^2 + 1206^2 = 1828980 = 36 \times 50805$; for (v), each of $45^2 + 5005^2$,

$2967^2 + 4031^2$ and $3039^2 + 3977^2$ equals 25052050; and $92^2 + 27^2 = 9193$. A computer search confirms these, and finds that 25×1002082 actually has six representations with $0 < r < s$ (the other three are $(721, 4953)$, $(2079, 4553)$ and $(2395, 4395)$). They come from a second representation of 1002082 itself, so any three distinct pairs are acceptable. Similarly $9193 = 48^2 + 83^2$ as well.

Common slips

- In (v), counting $(3 + 4\mathrm{i})$ and $(4 + 3\mathrm{i})$ as giving different pairs: they give the same pair in the opposite order.
- In (vi), dividing by the wrong conjugate and getting a non-integer, then giving up. Try the other one.
- Sign errors in $\mathrm{i}^2 = -1$ when expanding products: always check the final sum of squares directly.

Beyond STEP

The numbers $a + b\mathrm{i}$ with a, b integers form the *Gaussian integers*, a ring in which, like the ordinary integers, factorisation into primes is unique. An ordinary prime $p \equiv 1 \pmod 4$ splits as $(x + y\mathrm{i})(x - y\mathrm{i})$, which is Fermat's two-squares theorem. Counting the ways of sharing these factors out explains exactly how many representations a number has, and why 1002082 has more than the one we used. Davenport's *The Higher Arithmetic* treats sums of two squares beautifully; Allenby's *Rings, Fields and Groups* covers the Gaussian integers as a ring.

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